

AI Agents

Prepared for KPMG

The AI Agents course is an enhanced hands-on training experience focused on the practical application of AI agents within a business context. It covers essential topics such as Agentic AI tools, business use cases, process re-engineering with agents, and modern AI agent tool implementations, including Pinecone, LangChain, LangSmith, and Llama Index. The course emphasizes learning by doing with interactive labs and exercises. It is designed for those seeking to understand, implement, and strategically apply AI agent technologies in real-world business scenarios.

Audience

This course is ideal for:

- Developers
- Data Scientists
- Technical Managers

Prerequisites

In order to succeed in this course, participants will need:

- Basic Python programming experience
- Understanding of databases and data concepts
- AI/ML fundamentals

Objectives

After this course, participants will be able to:

- Understand the fundamentals of AI agents and their business applications
- Apply key AI agent tools such as Pinecone, LangChain, LangSmith, and Llama Index.
- Analyze and re-engineer business processes using AI agents.
- Compare and contrast common agent designs
- Execute AI-generated code safely
- Develop AI agents for specific business applications
- Utilize various AI agent tools

Length

2 days

Outline

Introduction to AI Agents in Business

- What are AI agents?
- Business value and impact of AI agents
- Overview of Agentic AI tools
- Business use cases for AI agents across industries
- Practical exercises: Identifying potential agent applications in specific business contexts

Agentic AI Tools and Modern Implementations

- Introduction to modern tools in AI agents (Pinecone, LangChain, LangSmith, Llama Index)
- Tool selection and integration strategies
- Deep dive into tool functionalities and use cases
- Practical exercises: Setting up and using Pinecone, LangChain, LangSmith, and Llama Index

Common Agent Designs

- Dedicated sub-agents architecture
- Agent networks and communication
- Hybrid agent designs
- Use cases and trade-offs for different architectures
- Case study: Real-world agent architectures

Executing Generated Code

- Security considerations for code execution
- Sandboxing and controlled environments
- Error handling and debugging
- Best practices for safe code generation and execution

AI Agents for Business Applications

- Agent for database search and chat for customer support
 - Architecture and workflow for business context
 - Integration with business systems
 - User interface and interaction for customer service
- Agent for data analysis and chart creation for business intelligence
 - Data processing and transformation for business data
 - Chart generation for business reporting
 - Visualization and interpretation for business insights
- Other example agents (if time allows)

Process Re-engineering with AI Agents

- Identifying processes suitable for agent implementation
- Steps in re-engineering processes with AI agents
- Measuring the impact of agent implementation
- Discussion: Challenges and solutions in process re-engineering